Tourism Experience Analytics

Project Report

Data Science & Machine Learning

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Domain	Tourism & Travel Analytics
Tools	Python, Scikit-learn, XGBoost, LightGBM
App	Streamlit Cloud
GitHub	tourism-analytics

Abstract

Tourism is one of the fastest-growing industries worldwide, generating vast amounts of user interaction data that can be utilized to improve traveler experiences and business decision-making. This project presents a Tourism Experience Analytics: Classification, Prediction, and Recommendation System, which leverages data analytics, machine learning, and recommendation techniques to analyze tourist behavior and provide intelligent, personalized travel insights. The system is developed using a relational tourism dataset consisting of 52,930 tourist transactions involving 33,530 users and 30 attractions across multiple continents, covering travel activities from 2013 to 2022. The dataset includes nine interconnected tables containing transaction history, user demographics, attraction information, and geographical reference data.

The project addresses three major machine learning objectives. First, a regression model was developed to predict attraction ratings based on user behavior and attraction characteristics. Among multiple algorithms tested, Linear Regression achieved the best performance with an R^2 score of 0.7375 and a Mean Absolute Error (MAE) of 0.29. Second, a multi-class classification model was designed to predict users' travel modes, including Couple, Family, Friends, Solo, and Business travel categories. After handling class imbalance using SMOTE, XGBoost delivered the best performance with a weighted F1-score of 0.4565. Third, a hybrid recommendation system combining collaborative filtering and content-based filtering was implemented to generate personalized tourist attraction recommendations.

The project further incorporates data cleaning, preprocessing, feature engineering, exploratory data analysis (EDA), visualization, and model evaluation to ensure reliable and interpretable outputs. All predictive models and recommendation components were integrated into a publicly deployed Streamlit web application, providing users with interactive dashboards, attraction rating prediction, visit mode prediction, and personalized attraction recommendations through a user-friendly interface.

This project demonstrates how machine learning and analytics can transform tourism services by enabling intelligent decision-making, improving customer experience, and delivering personalized travel recommendations through data-driven solutions.

1. Introduction

1.1 Introduction

The tourism and hospitality industry is one of the largest and fastest-growing sectors in the global economy, attracting billions of travelers to destinations worldwide each year. With the expansion of digital travel platforms, online booking systems, and user-generated content, enormous amounts of tourism-related data are continuously being generated through customer interactions, reviews, ratings, bookings, and travel activities. Effectively utilizing this data provides opportunities to better understand tourist behavior, preferences, and satisfaction, enabling organizations to deliver more personalized and engaging travel experiences.

Traditional tourism recommendation systems commonly rely on popularity-based approaches or broad demographic segmentation, which often fail to capture the unique preferences and travel patterns of individual users. Recent advancements in data science and machine learning provide powerful alternatives by identifying hidden patterns within historical tourism data and generating personalized predictions and recommendations at scale.

This project, **"Tourism Experience Analytics: Classification, Prediction, and Recommendation System,"** aims to leverage machine learning and analytics techniques to build an intelligent tourism decision-support platform. The system focuses on three primary objectives. First, it predicts the rating a tourist is likely to assign to an attraction using historical behavior and contextual information. Second, it predicts the tourist's visit mode, identifying whether the individual is traveling as a **Couple, Family, Friends group, Solo traveler, or Business traveler**. Third, it generates personalized attraction recommendations through a hybrid recommendation approach that combines **collaborative filtering** and **content-based filtering** techniques.

The project follows a complete data science workflow, beginning with **data collection, preprocessing, cleaning, exploratory data analysis (EDA), feature engineering, model development, and evaluation**, followed by deployment into an interactive application environment. The dataset contains **52,930 tourist transactions involving 33,530 users and 30 attractions**, spanning the years **2013 to 2022**, and consists of nine interconnected relational tables that capture transaction history, user demographics, attraction information, and geographical reference details.

To make the system accessible and practical for real-world use, all predictive and recommendation components are integrated into a **Streamlit web application** and deployed through **Streamlit Community Cloud**. The application enables users to explore tourism analytics, predict attraction ratings, estimate visit modes, and receive personalized attraction recommendations through an intuitive interface.

This project demonstrates how machine learning and recommendation systems can be effectively applied within the tourism domain to improve user experience, enhance personalization, and support data-driven decision-making for tourism businesses and digital travel platforms.

1.2 Problem Statement

The tourism industry generates large volumes of data through user visits, ratings, and travel activities, but many platforms struggle to utilize this information effectively for personalization and decision-making. Traditional recommendation methods often rely on popularity-based approaches, which fail to capture individual traveler preferences and behavior patterns.

This project addresses key challenges including the lack of personalized attraction recommendations, inability to predict tourist satisfaction, limited understanding of visit behavior, underutilization of relational tourism data, and class imbalance in visit mode prediction.

To solve these problems, the project develops an end-to-end tourism analytics system that integrates relational data, performs preprocessing and feature engineering, builds machine learning models for rating prediction and visit mode classification, and implements a hybrid recommendation system. The complete solution is deployed as a Streamlit web application to provide tourism analytics, predictive insights, and personalized attraction recommendations.

1.3 Objectives

- **Data Cleaning** – Handle missing values, duplicates, and data inconsistencies across all relational tables to ensure high-quality data for analysis.
- **Data Integration** – Merge multiple relational tables into a unified master dataset and engineer meaningful features such as user average rating, attraction visit count, and seasonal visit patterns.
- **Exploratory Data Analysis (EDA)** – Analyze tourist behavior, rating distributions, geographic trends, visit patterns, and class imbalance using statistical analysis and visualizations.
- **Regression Modeling** – Develop and compare machine learning models to predict tourist attraction ratings using evaluation metrics such as MAE, RMSE, and R^2 score.
- **Classification Modeling** – Build predictive models to classify tourist visit modes while addressing class imbalance using SMOTE and evaluating performance through Accuracy and F1-score.
- **Recommendation System Development** – Design a hybrid recommendation system combining collaborative filtering and content-based filtering to generate personalized attraction suggestions.
- **Web Application Deployment** – Develop and deploy an interactive Streamlit application integrating predictive models, recommendation systems, and tourism analytics dashboards.
- **Documentation and Reporting** – Document methodologies, model performance, analytical findings, and project outcomes in a comprehensive project report.

2. System Design and Dataset Information

2.1 Tech Stack

Category	Tools Used
Language	Python 3.12
Data Processing	Pandas, NumPy
Visualization	Matplotlib, Seaborn
Machine Learning	Scikit-learn, XGBoost, LightGBM
Imbalance Handling	imbalanced-learn (SMOTE)
Development	Google Colab, VS Code
Deployment	Streamlit Cloud
Version Control	GitHub

2.2 Dataset Description

The dataset used in this project is a relational tourism dataset consisting of nine interlinked tables that capture tourist visit behavior, user demographics, and attraction details across the world.

2.3 Dataset Overview

Table	Rows	Columns	Description
Transaction	52,930	7	Core visit records with ratings
User	33,530	5	User demographic information
City	9,143	3	City reference data
Country	165	3	Country reference data
Region	22	3	Regional groupings
Continent	6	2	Continental groupings
Updated Item	1,698	5	Attraction details
Type	17	2	Attraction type categories
Mode	6	2	Visit mode labels

2.4 Key Statistics

- Total transactions: 52,930
- Unique users: 33,530
- Unique attractions visited: 30 out of 1,698 available
- Visit years: 2013 to 2022
- Average rating: 4.16 out of 5
- Most common visit mode: Couples (40%)

2.5 Table Description

Table	Row	Column	Description
Transaction	52,930	7	Core table — each row is a tourist visit with rating, year, month, visit mode and attraction
User	33,530	5	Tourist demographics — continent, region, country and city of each user
Item (Updated)	1,698	5	Attraction details — name, address, type and city of each attraction
City	9,142	3	Reference table mapping city IDs to city names
Country	165	3	Reference table mapping country IDs to country names
Region	22	3	Reference table mapping region IDs to region names
Continent	6	2	Reference table mapping continent IDs to continent names
Type	17	2	Reference table listing 17 attraction type categories
Mode	6	2	Reference table listing 5 visit mode categories

Visit Modes - Business, Couples, Family, Friends, Solo

Attraction Types - Beaches, National Parks, Religious Sites, Volcanos, Water Parks, Waterfalls, Historic Sites, Museums and more

3. Implementation and Analysis

3.1 Data Cleaning

The data cleaning phase focused on improving data quality by handling missing values, validating duplicates, and resolving data type inconsistencies across all nine relational tables. The dataset was largely clean, containing only **5 missing values** across all tables.

3.1.1 Null Value Handling

Missing value analysis was performed across all nine relational tables to identify incomplete or inconsistent records. The dataset was found to be highly clean, containing only five missing values across all tables.

Among these missing values, one record was identified in the `CityName` column of the City table. Since the corresponding CityId did not have any associated users, the record was removed from the dataset. Additionally, four missing values were identified in the `CityId` column of the User table. These missing entries were replaced with **0**, representing an unknown city category.

Handling missing values at the preprocessing stage ensured better data consistency, improved dataset quality, and reduced the possibility of errors during feature engineering, model training, and analysis.

3.1.2 Data Type Correction

Data type validation was performed to ensure consistency across all relational tables before feature engineering and model development. During this process, an inconsistency was identified in the `AttractionTypeId` column of the Item table. Some records contained numeric identifiers, while others stored textual category values such as **"Beach"**, **"Temple"**, and **"Park"**.

To standardize the data, a mapping approach was implemented using the Type reference table, converting textual attraction categories into their corresponding numeric identifiers. This ensured uniform representation of attraction types throughout the dataset and prevented potential issues during data integration, encoding, and machine learning model training.

3.1.3 Duplicate Check

Duplicate validation was performed across all nine relational tables to ensure data reliability and eliminate redundant records before further analysis. Duplicate checks were conducted at both the **row level** and **primary key level** to verify data uniqueness.

The validation confirmed that no duplicate entries were present in the dataset. Additionally, all primary identifiers, including **TransactionId, UserId, CityId, and AttractionId**, were found to be unique and consistent across their respective tables.

This verification ensured data integrity and provided a clean foundation for feature engineering, exploratory data analysis, machine learning model development, and recommendation system implementation.

3.2 Data Merging and Feature Engineering

3.2.1 Table Merging

After completing the data cleaning process, all nine relational tables were combined into a single master dataset to create a unified dataset for analysis and machine learning tasks. The **Transaction** table was selected as the primary table because it contained the core tourist interaction information.

The remaining tables, including **Visit Mode, Attraction Details, Attraction Type, User Information, City, Country, Region, and Continent**, were merged sequentially using **left join operations**. This approach ensured that all tourist transaction records were retained while enriching them with additional demographic, geographic, and attraction-related information.

After merging, redundant columns generated during joins were removed, and column names were standardized to improve readability and maintain consistency across the dataset.

3.2.2 Feature Engineering

Feature engineering was performed to create additional variables that could improve model performance and capture important tourist behavior patterns.

The **User_Avg_Rating** feature was created to represent the average rating provided by a user across all attraction visits. This feature helps understand individual user preferences and rating behavior.

The **User_Visit_Count** feature was introduced to measure the total number of visits made by each user, providing insight into user engagement and travel activity levels.

Attraction-related behavioral features were also developed. The **Attraction_Avg_Rating** feature captures the average rating received by an attraction, reflecting its overall popularity and visitor satisfaction. The **Attraction_Visit_Count** feature measures the total number of visits associated with each attraction.

In addition, a **Season** feature was derived from the visit month information by categorizing visits into **Spring, Summer, Autumn, and Winter**. This feature was created to capture potential seasonal tourism patterns and travel trends.

3.2.3 Master Dataset

After completing data merging and feature engineering, the final master dataset contained **52,930 records and 25 columns** with **zero missing values**. The dataset was fully prepared for subsequent stages including Exploratory Data Analysis (EDA), machine learning model development, prediction tasks, and recommendation system implementation.

3.3 Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was performed on the final master dataset to understand tourist behavior patterns, identify important trends, analyze feature relationships, and generate insights that could support machine learning model development.

3.3.1 Rating Distribution

Tourist ratings were analyzed to understand overall satisfaction levels across attraction visits. The rating distribution showed a strong concentration toward higher ratings. Approximately **79% of all ratings were either 4 or 5 stars**, indicating that tourists generally had positive experiences during their visits. The average tourist rating across the dataset was **4.16 out of 5**, demonstrating an overall high satisfaction level. This observation suggests that most attractions received favorable feedback, making rating prediction an important component of the project.

3.3.2 Visit Mode Distribution

The distribution of tourist visit modes was examined to understand travel behavior patterns. The analysis showed that **Couples travel** represented the largest segment, accounting for approximately **40% of total visits**, followed by **Family travel (29%)** and **Friends travel (21%)**. Solo travel contributed a smaller proportion, while **Business travel represented only 1% of total visits**. This distribution revealed a significant class imbalance problem in the visit mode classification task. To address this issue during model development, **SMOTE (Synthetic Minority Oversampling Technique)** was applied to improve minority class representation.

3.3.3 Attraction Popularity Analysis

Tourist attraction visit frequencies were analyzed to identify highly visited locations. The analysis indicated that tourist activity was concentrated around attractions located in **Indonesia, particularly Bali**. The **Sacred Monkey Forest Sanctuary** emerged as the most visited attraction in the dataset. These findings demonstrate location-based tourist concentration and attraction popularity trends.

3.3.4 Geographic Distribution of Visits

Tourist visits were analyzed geographically across continents. The analysis showed that **Asia and Europe contributed the highest tourist activity**, indicating strong travel engagement within these regions. America and Oceania contributed moderate tourist volume, while Africa showed comparatively lower representation in the dataset. This geographic trend provides insight into tourism demand patterns across global regions.

3.3.5 Seasonal Analysis

Tourist ratings were examined across different seasons to understand whether seasonal variation influenced visitor satisfaction. The results indicated that ratings remained relatively stable across **Spring, Summer, Autumn, and Winter**, suggesting that seasonal changes had minimal influence on tourist satisfaction levels. This finding indicates that attraction quality and user preferences contribute more significantly to rating behavior than seasonal timing.

3.3.6 Attraction Type Analysis

Different attraction categories were analyzed to identify tourist preferences. The analysis revealed that **Beaches, Nature and Wildlife Areas, Points of Interest and Landmarks, Religious Sites, and National Parks** represented the most popular attraction categories. The findings suggest a strong tourist preference toward outdoor experiences, scenic locations, and cultural destinations.

3.3.7 Correlation Analysis

Correlation analysis was conducted to identify relationships between numerical features and determine important predictors for machine learning models. The strongest positive relationships with tourist ratings were observed for **Attraction_Avg_Rating** and **User_Avg_Rating**, indicating that attraction quality and user rating behavior strongly influence rating prediction. Features such as **Visit Month** and **Visit Year** showed relatively weak relationships with ratings, suggesting limited predictive importance. These findings helped guide feature selection during machine learning model development.

The EDA process provided important insights into tourist satisfaction patterns, travel behavior, attraction popularity, and feature relationships, forming a strong foundation for predictive modeling and recommendation system development.

4. Model & Deployment

4.1 Regression - Predict Rating

Three regression models were trained to predict the rating a user will give to an attraction based on their demographics and visit details.

Objective: Predict the rating (1–5) a tourist will give to an attraction based on their profile and visit details.

Features Used: VisitYear, VisitMonth, AttractionTypeId, ContinentId, RegionId, CountryId, User_Avg_Rating, User_Visit_Count, Attraction_Avg_Rating, Attraction_Visit_Count

Train/Test Split: 80% training — 42,344 samples | 20% testing — 10,586 samples

Results:

Model	MAE	RMSE	R2 Score
Linear Regression	0.2903	0.4972	0.7375 (Best)
XGBoost	0.2719	0.5020	0.7324
Random Forest	0.2759	0.5414	0.6888

Best Model: Linear Regression with R^2 of 0.7375 predicts ratings with an average error of only 0.29 on a 1- 5 scale.

4.2 Classification - Predict Visit Mode

Three classification models were trained to predict how a user travels. Class imbalance was handled using SMOTE (Synthetic Minority Oversampling Technique) which balanced all 5 classes to 17,296 samples each in the training set.

Objective: Predict how a tourist travels - Couples, Family, Friends, Solo or Business.

Features Used: VisitYear, VisitMonth, AttractionId, AttractionTypeId, ContinentId, RegionId, CountryId, Rating, User_Avg_Rating, User_Visit_Count, Attraction_Avg_Rating, Attraction_Visit_Count

Class Imbalance Handling: SMOTE applied - all 5 classes balanced to 17,296 samples each in training data.

Train/Test Split: 80% training | 20% testing

Results:

Model	Accuracy	F1 Score	Precision	Recall
XGBoost	0.4623	0.4565 (Best)	0.4652	0.4623
LightGBM	0.4590	0.4483	0.4648	0.4590
Random Forest	0.4451	0.4492	0.4546	0.4451

Best Model: XGBoost with F1 score of 0.4565. Couples and Family modes predicted most accurately due to higher data representation.

4.3 Recommendation System

Objective: Suggest personalized attractions to tourists based on their visit history.

Collaborative Filtering:

- Computes cosine similarity between users based on their rating patterns
- Finds top 50 similar users and recommends attractions they visited but the target user has not
- Fallback to top rated attractions overall when no similar user data is available

Content Based Filtering:

- Builds attraction feature matrix using AttractionTypeId, Attraction_Avg_Rating and Attraction_Visit_Count
- Normalizes features using MinMaxScaler
- Computes cosine similarity between attractions
- Scores unvisited attractions by weighted similarity to previously visited attractions

Hybrid System: Both approaches are combined to provide comprehensive recommendations collaborative filtering captures user behavior patterns while content based filtering captures attraction feature similarity.

4.4 Streamlit Web Application

Objective: Deploy all models into an accessible interactive web application.

Page	Features
Home	Key metrics — total visits, users, attractions, average rating
EDA Dashboard	Interactive charts across Ratings, Visits and Geography tabs
Predict Rating	User inputs profile → predicts expected attraction rating

Predict Visit Mode	User inputs details → predicts travel mode
Get Recommendations	User inputs ID → collaborative + content based recommendations

Deployment Details:

App URL - https://tourism-analytics-lspgujljb57fg6tfgdkxtg.streamlit.app/

5. Results & Conclusion

5.1 Key Insights

The exploratory analysis and machine learning modeling process revealed several important findings regarding tourist behavior, attraction preferences, and model performance.

1. **Tourist Satisfaction is Generally High**

   Tourist ratings were strongly concentrated toward higher values, with an overall average rating of 4.16 out of 5. Approximately 79% of all attraction ratings were either 4 or 5 stars, indicating that tourists are generally satisfied with their travel experiences and attraction visits.

2. **Couples Represent the Largest Tourist Segment**

   Travel behavior analysis showed that Couples travel accounts for nearly 40% of all visits, making it the dominant travel category within the dataset. Family travel represents the second-largest segment, while business travel contributes only a very small proportion of total tourist activity. This insight highlights the importance of considering travel group preferences when building tourism recommendation systems.

3. **Seasonal Effects Have Limited Influence on Ratings**

   Seasonal analysis revealed that tourist ratings remain relatively stable across Spring, Summer, Autumn, and Winter. Correlation analysis further confirmed that visit month has minimal influence on tourist satisfaction.This suggests that attraction quality and visitor preferences play a more important role than seasonal timing.

4. **Attraction Quality Strongly Influences Tourist Ratings**

   Feature relationship analysis identified Attraction_Avg_Rating and User_Avg_Rating as the strongest predictors of tourist ratings. Tourists with historically higher rating tendencies generally continue exhibiting similar behavior, while highly rated attractions consistently receive positive feedback. These strong relationships contributed significantly to model performance.

5. **Outdoor Attractions Drive Tourism Activity**

   Tourist attraction analysis showed that Beaches, Nature and Wildlife Areas, National Parks, and Scenic Locations dominate attraction visits. The dataset also revealed that Indonesia, particularly Bali, represents one of the most frequently visited tourism regions, with Sacred Monkey Forest Sanctuary emerging as the most visited attraction.

6. **Class Imbalance Significantly Impacts Classification**

   Visit mode analysis revealed substantial imbalance across tourist categories. Couples represented approximately 40% of visits, while Business travel accounted for only 1%, creating challenges for classification model development. To improve minority class prediction performance, SMOTE (Synthetic Minority Oversampling Technique) was applied during training.

7. **Recommendation Quality Improves with Rich User History**

   The recommendation system demonstrated stronger performance when sufficient user interaction history was available. Users with limited visit records relied more heavily on fallback recommendations, while users with richer historical interactions received more personalized attraction suggestions.

8. **Feature Engineering Contributed Significantly to Model Performance**

   Engineered features such as User_Avg_Rating and Attraction_Avg_Rating proved highly influential in improving predictive performance. The results further demonstrated that simpler machine learning models can outperform more complex approaches when meaningful feature engineering and strong feature relationships exist within the data.

These insights provided valuable understanding of tourist behavior patterns and directly contributed to improving prediction accuracy, recommendation quality, and overall system effectiveness.

5.2 Recommendations

Improve Dataset Quality: Include additional features such as age, gender, budget, trip duration, and travel frequency to improve prediction performance.

Improve Classification Performance: Collect more data for underrepresented classes such as Business and Solo travelers to reduce class imbalance.

Enhance Recommendation System: Implement advanced recommendation techniques such as **SVD** and location-based recommendations to improve personalization.

Improve Rating Prediction: Include attraction-specific information like ticket price, accessibility, and facilities to improve regression accuracy.

Expand Web Application: Add features such as user login, real-time tourism data integration, and interactive maps.

Improve Production Deployment: Implement automated model retraining, monitoring systems, and database integration for scalability.

These improvements can further enhance prediction accuracy, recommendation quality, and overall user experience.

5.3 Conclusion

This project successfully developed an end-to-end **Tourism Experience Analytics: Classification, Prediction, and Recommendation System**, demonstrating the practical application of data science and machine learning in the tourism domain. The project followed a complete pipeline including **data cleaning, data integration, feature engineering, exploratory data analysis, model development, evaluation, and deployment**.

The dataset containing **52,930 tourist transactions across 33,530 users and 30 attractions** was processed and transformed into a machine-learning-ready master dataset. Feature engineering played a major role in improving predictive performance by introducing behavioral and attraction-related features.

The regression model achieved an **R^2 score of 0.7375**, accurately predicting tourist ratings, while the classification model predicted visit modes after handling class imbalance using **SMOTE**. A hybrid recommendation system combining **collaborative filtering and content-based filtering** was also implemented to provide personalized attraction suggestions.

All components were integrated into an interactive **Streamlit web application**, allowing users to explore tourism insights, predict attraction ratings, estimate visit modes, and receive personalized recommendations.

The project highlights that effective feature engineering and data-driven approaches can significantly improve tourism analytics, enabling better tourist satisfaction prediction, personalized recommendations, and intelligent decision-making for tourism platforms.

5.4 Future scope

Real-Time Data Integration — Connect to live tourism APIs for dynamic attraction data and automatic model updates as new visit records are added.

Sentiment Analysis — Analyze tourist text reviews using NLP to extract sentiment scores as additional features for improved predictions.

Mobile Application — Develop a mobile version for iOS and Android enabling tourists to access recommendations and predictions during their travels.

Map-Based Visualization — Integrate interactive maps to display attraction locations and help tourists plan travel routes geographically.

Social Features — Enable tourists to share experiences and implement group recommendation features for trip planning together.